

Business Requirements Document

E-Commerce Sales & Customer Analytics

Personal Portfolio Project — Data Analytics

Project Type:	End-to-End Data Analytics (Portfolio)
Data Sources:	3 CSV datasets — Customers, Products, Orders
Dataset Scale:	25,000 customers 1,175 products 138,116 orders
Time Coverage:	Jan 1, 2021 – Dec 31, 2025
Target Stack:	SQL (data prep & analysis) + Power BI / Tableau (dashboards)
Document Version:	1.0
Status:	Draft / Proposed

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1. Project Overview

This project is a self-directed, end-to-end data analytics portfolio piece built around a realistic e-commerce dataset. It covers the full analytics lifecycle: loading raw data into a relational database, cleaning and validating it with SQL, and building an interactive Power BI (or Tableau) dashboard that answers concrete business questions about sales, products, and customers.

The goal is to demonstrate the skills a data analyst uses on the job: writing structured SQL queries, reasoning about data quality, designing a usable data model, defining KPIs that matter to a business, and communicating findings through dashboards and a short written summary of insights.

The finished solution should let a viewer answer questions such as:

- How is revenue and profit trending over the last five years?
- Which customer segments, regions, and countries generate the most value?
- How healthy is the order pipeline — completion, cancellation, and return rates?
- Which marketing channels and payment methods are associated with the most orders?
- How satisfied are customers, based on ratings and review sentiment?
- Who are the most valuable and most loyal customers?

2. Business Problem

The dataset represents an e-commerce business with five years of order history (2021–2025), a base of 25,000 customers, and a catalog of 1,175 products. On its own, this raw transactional data cannot answer management-level questions directly — it needs to be cleaned, modeled, and aggregated before it becomes useful.

Framed as a business problem, stakeholders would want a centralized analytics solution that answers questions such as:

- What is total revenue, net revenue, and profit, and how are they trending?
- Which regions, countries, and customer segments drive the most revenue?
- What share of orders are completed, cancelled, pending, or returned, and why?
- Are discounts eroding profit margin on certain orders or segments?
- Which payment methods, sales channels, and shipping methods are most used?
- How reliable is order fulfillment — on-time delivery rate, average delivery days?
- Which marketing channels and campaigns correlate with the most orders?
- How satisfied are customers, and does sentiment differ by channel or region?
- Who are the highest-value, most loyal customers, and how repeat-heavy is the business?

This project treats those questions as the requirements the analytics build must satisfy.

3. Project Objectives

- Load the three source CSV files into a structured relational database (SQL Server / PostgreSQL / MySQL) with appropriate data types and constraints.

- Profile and document data quality: duplicates, missing values, and referential integrity between customers and orders.
- Write SQL queries covering aggregation, joins, filtering, grouping, subqueries, CTEs, CASE logic, and window functions to answer the business questions in Section 2.
- Define and calculate a consistent set of sales, customer, and operational KPIs.
- Build a Power BI (or Tableau) data model connecting the cleaned tables.
- Design interactive dashboards covering executive, sales, customer, and operations views.
- Summarize key findings, insights, and recommendations as a business-facing deliverable.
- Document assumptions, limitations, and possible future extensions of the project.

4. Stakeholders

As a personal project, there is no live business audience — the roles below are used as design personas to keep the analysis grounded in realistic decision-making needs, and double as the audience this project is meant to demonstrate value to (recruiters, hiring managers, and reviewers).

Stakeholder Persona	Primary Interest
Business / General Manager	Overall performance: revenue, profit, growth trend
Sales Manager	Sales trends, order volume, channel and payment mix
Product / Category Manager	Catalog structure, pricing, category and brand composition
Marketing Manager	Channel performance, campaigns, coupon usage, acquisition cost
Operations / Logistics Manager	Delivery performance, warehouse load, returns
Customer Experience Lead	Ratings, review sentiment, satisfaction by segment
Data Analyst (project owner)	Data preparation, analysis, dashboard design, documentation

5. Data Sources

The project uses three flat files, each representing one entity in the analytics model.

File	Rows	Cols	Contents
customers.csv	25,000	11	One row per customer: demographics, segment, and location
products.csv	1,175	9	Product catalog: category, brand, supplier, pricing
orders.csv	138,116	37	One row per order: status, fulfillment, financials, marketing, reviews, and loyalty detail

■ Key structural limitation

orders.csv does not contain a product_id or line-item table linking an order to a specific product. Order-level analysis (revenue, profit, customer behavior) and product-catalog analysis (pricing, category mix) can each be done in full, but revenue cannot currently be attributed to individual products or categories. This is carried through the rest of the document as a documented scope boundary and a candidate future enhancement (see Sections 6.2 and 16).

6. Scope

6.1 In Scope

Sales & Financial Performance

- Gross revenue, net revenue, discounts, tax, shipping cost, and profit
- Profit margin percentage and loss-making order analysis
- Order volume and Average Order Value (AOV)
- Sales trends by day, month, quarter, and year (2021–2025)
- Sales by sales channel, payment method, and currency tag

Order Fulfillment & Logistics

- Order status mix: Completed, Returned, Cancelled, Pending
- Delivery performance: on-time / delayed / early / cancelled, average delivery days
- Shipping method and warehouse-level order load (19 warehouses)
- Return rate and return reasons (where recorded)

Customers

- Customer counts, new vs. returning vs. loyal customer mix
- Customer segmentation (Consumer, Premium, Business, VIP)
- Customer lifetime value, order frequency, repeat-purchase rate
- Customer satisfaction: rating and review sentiment (completed orders only)

Geography

- Sales and customer distribution by country (7 countries), region (5), state, and city
- Customer acquisition cost by segment / region

Marketing & Loyalty

- Order volume and value by marketing channel (10 channels) and campaign
- Coupon usage and its association with order value
- Loyalty points earned and redeemed

Product Catalog (standalone)

- Category, sub-category, brand, and supplier composition of the catalog

- Pricing and margin structure by category and brand (catalog-level, not order-linked)
- Product rating distribution

6.2 Out of Scope

- Product- or category-level revenue attribution (no order-to-product key exists in the source data — see Section 5)
- Inventory levels and stock-out / restocking analysis
- Real-time or streaming analytics
- Fraud detection
- Product recommendation systems / machine learning models
- Delivery route optimization
- Currency conversion modeling beyond the assumption documented in Section 13

These can be considered future enhancements (Section 16).

7. Data Dictionary

7.1 Customers (customers.csv — 25,000 rows)

Field	Type	Description
customer_id	string	Unique customer identifier (e.g. CUST-000001)
customer_name	string	Customer full name
customer_age	integer	Customer age (observed range: 18–74)
gender	string	Female / Male / Non-Binary
customer_segment	string	Consumer / Premium / Business / VIP
customer_city	string	Customer city
customer_state	string	Customer state or province
customer_country	string	7 distinct values: UAE, Germany, USA, UK, India, Canada, Australia
region	string	South / North / West / East / Central
customer_postal_code	integer	Postal code
customer_acquisition_cost	float	Estimated cost to acquire the customer

7.2 Products (products.csv — 1,175 rows)

Field	Type	Description
product_id	string	Unique product identifier (e.g. PROD-000001)
product_name	string	Product name
product_category	string	15 categories, e.g. Electronics, Fashion, Grocery
product_subcategory	string	117 distinct sub-categories
brand	string	144 distinct brands
supplier	string	10 distinct suppliers
unit_price	float	Product selling price
product_cost	float	Product cost to the business
product_rating	float	Average product rating

7.3 Orders (orders.csv — 138,116 rows)

The orders table is the widest and richest of the three, covering order status, fulfillment, financials, marketing attribution, loyalty, and post-purchase feedback in a single row per order.

Field(s)	Type	Description
order_id	string	Unique order identifier (e.g. ORD-301242)
order_date / order_time	date / time	Date and time the order was placed (2021–2025)
order_status	string	Completed / Returned / Cancelled / Pending
sales_channel	string	Website / Mobile App / Marketplace / Social Media
customer_id	string	Foreign key to customers.customer_id
customer_type	string	New / Returning / Loyal
payment_method	string	7 methods, e.g. Credit Card, Digital Wallet, PayPal
payment_status	string	Paid / Pending / Failed / Refunded
currency	string	Currency tag on the order (7 values — see Section 13 assumption)
shipping_method	string	Standard / Express / Economy / Same Day
warehouse	string	Fulfilling warehouse (19 distinct warehouses, WH-001 – WH-019)
delivery_days / estimated_delivery_days	float	Actual vs. estimated delivery time (populated for completed orders only)

Field(s)	Type	Description
delivery_status	string	On Time / Delayed / Early / Cancelled
return_status / return_reason	string	Populated only when order_status = Returned
customer_rating / review_sentiment / customer_review	float / string / string	Post-purchase feedback (completed orders only)
marketing_channel / campaign_name / coupon_code	string	Attribution and promotion detail (campaign and coupon fields are blank when not applicable)
loyalty_points_earned / loyalty_points_redeemed	integer	Loyalty program activity
quantity	integer	Units purchased in the order (observed range: 1–24)
gross_sales / discount_amount / tax_amount / shipping_cost / net_sales	float	Order-level financial breakdown
product_cost / profit / profit_margin_percentage	float	Cost and profitability at the order level
customer_lifetime_value	float	Running customer lifetime value at time of order
is_repeat_customer / customer_order_count	boolean / integer	Repeat-purchase flag and running order count for the customer

8. Data Quality Assessment

A profiling pass was run over all three files before analysis. The findings below should be documented in the project's data-cleaning notes and, where relevant, handled explicitly in SQL (e.g. with COALESCE, CASE, or filtered joins) rather than silently dropped.

8.1 Completeness & Uniqueness

Field(s)	Finding
customer_id / product_id / order_id	0 duplicates found in any of the three files
customers.csv / products.csv	No missing values in any column
orders.customer_id	Every order's customer_id exists in customers.customer_id (0 orphan rows) — referential integrity is clean
delivery_days, estimated_delivery_days, customer_rating, review_sentiment, customer_review	Blank on ~24,557 rows — exactly the orders that are Cancelled, Pending, or Returned. This is expected: undelivered orders have no delivery time or review, not a data error
return_status / return_reason	Blank on ~128,654 rows — exactly the orders that are not Returned. Expected, not an error
campaign_name / coupon_code	Blank on ~83,233 and ~110,502 rows respectively, corresponding to orders with no associated campaign or coupon

8.2 Business Rule Checks

- No negative values found in quantity, gross_sales, or net_sales.
- 1,296 orders (about 0.9%) show negative profit — legitimate loss-making orders, likely driven by heavy discounting; worth a dedicated dashboard view rather than treating as an error.
- order_status and return_status are consistent: return_status is populated if, and only if, order_status = Returned.
- is_repeat_customer and customer_order_count agree throughout the sample checked (order counts of 1 correspond to is_repeat_customer = False, and vice versa).

■ Assumption requiring confirmation

The currency field carries 7 distinct currency codes (USD, EUR, GBP, AED, AUD, CAD, INR), but the average order value is statistically similar across all currency groups (roughly 1,370–1,395 per order regardless of currency). This suggests the monetary columns are already normalized to one reporting unit rather than expressed in local currency. This project treats all monetary fields as pre-normalized and does not apply currency conversion; this should be re-verified against the true source system before reuse in a real business context.

9. Key Performance Indicators (KPIs)

9.1 Sales & Profitability KPIs

KPI	Definition
Total Gross Revenue	SUM(gross_sales)
Total Net Revenue	SUM(net_sales)
Total Profit	SUM(profit)
Average Profit Margin %	AVG(profit_margin_percentage)
Total Orders	COUNT(DISTINCT order_id)
Average Order Value (AOV)	SUM(net_sales) / COUNT(DISTINCT order_id)
Total Units Sold	SUM(quantity)
Revenue Growth (period over period)	(Current Period Net Revenue – Prior Period Net Revenue) / Prior Period Net Revenue × 100
Discount Rate	SUM(discount_amount) / SUM(gross_sales)

9.2 Order & Fulfillment KPIs

- Order Status Mix — % Completed, Returned, Cancelled, Pending (COUNT per status / COUNT total)
- Return Rate — COUNT(order_status = 'Returned') / COUNT(all orders)
- Cancellation Rate — COUNT(order_status = 'Cancelled') / COUNT(all orders)
- On-Time Delivery Rate — COUNT(delivery_status = 'On Time') / COUNT(delivered orders)
- Average Delivery Days — AVG(delivery_days), for completed orders
- Orders per Warehouse — COUNT(order_id) grouped by warehouse

9.3 Customer KPIs

- Total Customers, New Customers, Returning Customers, Loyal Customers (by customer_type)
- Repeat Purchase Rate — COUNT(is_repeat_customer = TRUE) / COUNT(all customers with orders)
- Average Customer Lifetime Value — AVG(customer_lifetime_value)
- Average Orders per Customer — AVG(customer_order_count)
- Average Customer Rating — AVG(customer_rating), completed orders only
- Review Sentiment Mix — % Positive / Neutral / Negative

9.4 Marketing & Loyalty KPIs

- Orders by Marketing Channel and Campaign
- Coupon Usage Rate — COUNT(coupon_code IS NOT NULL) / COUNT(all orders)
- Average Customer Acquisition Cost, by segment and region
- Loyalty Points Issued vs. Redeemed — SUM(loyalty_points_earned) vs. SUM(loyalty_points_redeemed)

10. Business Requirements

BR-01 Sales & Profitability Analysis

The solution must report total and trended revenue, discounts, net revenue, and profit, and allow drill-down by day, month, quarter, and year across the full 2021–2025 range.

BR-02 Order & Fulfillment Analysis

The solution must break down orders by status, delivery performance, shipping method, and warehouse, and surface the return rate and cancellation rate as headline operational metrics.

BR-03 Customer Analysis

The solution must report customer counts by type and segment, repeat-purchase behavior, customer lifetime value, and satisfaction (rating and review sentiment).

BR-04 Geographic Analysis

The solution must allow revenue, order volume, and customer counts to be sliced by country, region, state, and city.

BR-05 Marketing & Loyalty Analysis

The solution must report order volume and value by marketing channel and campaign, coupon usage, acquisition cost, and loyalty point activity.

BR-06 Product Catalog Analysis

The solution must report catalog composition (category, sub-category, brand, supplier), pricing, and margin structure. As noted in Section 5, this analysis is catalog-level only and is not currently joinable to individual order transactions.

BR-07 Customer Segmentation

Where the data supports it, customers should be segmented by value and behavior — for example, high-value vs. low-value, frequent vs. one-time, and new vs. loyal — using `customer_lifetime_value`, `customer_order_count`, and `customer_type` as the primary inputs. A full RFM (Recency, Frequency, Monetary) model may be layered on top as an advanced feature.

11. SQL / Database Requirements

11.1 Database Development

- Create one table per source file (customers, products, orders) with explicit data types
- Define `customer_id` as the primary key on customers, and as a foreign key on orders
- Add appropriate constraints (NOT NULL on identifier fields, CHECK constraints on non-negative quantity / gross_sales)

- Note: because orders.csv has no product_id, no foreign-key relationship to products can be created without adding a synthetic mapping (see Section 16)

11.2 Data Preparation

- Standardize date/time fields into proper DATE and TIME (or DATETIME) types
- Explicitly handle NULLs in delivery, review, return, campaign, and coupon fields using CASE / COALESCE rather than treating them as errors (see Section 8.1)
- Validate referential integrity between orders and customers
- Flag and separately report the 1,296 negative-profit orders rather than excluding them

11.3 Analysis

SQL queries should demonstrate the following techniques against the KPIs and business requirements above:

- Aggregations (SUM, AVG, COUNT) and GROUP BY
- Multi-table JOINS (orders to customers)
- Filtering and CASE-based bucketing (e.g. order status groups, delivery buckets)
- Subqueries and CTEs for layered calculations (e.g. customer-level rollups feeding segment-level summaries)
- Window functions (e.g. running revenue, customer rank by lifetime value, month-over-month deltas)
- Date-based analysis (YEAR/MONTH extraction, period-over-period comparisons)

12. Power BI / Dashboard Requirements

Power BI (or an equivalent tool such as Tableau) will be used to build interactive dashboards on top of the cleaned data. Suggested global slicers: order date, country, region, customer segment, sales channel, and order status.

12.1 Page 1 — Executive Overview

KPI Cards

- Total Net Revenue
- Total Profit
- Total Orders
- Average Order Value
- Return Rate

Visuals

- Revenue and profit trend over time
- Order status mix (donut or stacked bar)
- Revenue by country / region
- Revenue by sales channel

12.2 Page 2 — Sales & Profitability

- Monthly / quarterly revenue and profit trend with drill-down to daily
- Discount rate and profit margin trend
- Revenue by payment method and sales channel
- Distribution and drivers of negative-profit orders

12.3 Page 3 — Customers

- New vs. returning vs. loyal customer counts over time
- Customer lifetime value distribution and top customers
- Repeat purchase rate and orders-per-customer distribution
- Customer rating and review sentiment breakdown

12.4 Page 4 — Operations & Fulfillment

- Order status funnel (Completed / Pending / Cancelled / Returned)
- On-time delivery rate and average delivery days, by shipping method and warehouse
- Orders by warehouse (load balancing view)
- Return rate and return reasons

12.5 Page 5 — Marketing & Loyalty

- Orders and revenue by marketing channel and campaign
- Coupon usage rate and its relationship to order value
- Customer acquisition cost by segment and region
- Loyalty points earned vs. redeemed over time

12.6 Page 6 — Product Catalog

- Category / sub-category / brand composition of the catalog
- Price and margin distribution by category
- Product rating distribution
- Explicitly labeled as catalog-level only (not linked to actual order transactions)

13. Assumptions

- Each customer, product, and order has a unique identifier, confirmed by the profiling in Section 8 (0 duplicates in any of the three files).
- All monetary fields in orders.csv are already expressed in a single normalized reporting unit; the currency field is retained for reference but is not used to convert amounts (see Section 8.2).
- Blank delivery, review, return, campaign, and coupon fields represent 'not applicable' rather than missing data, based on their consistent alignment with order_status.

- Negative-profit orders are valid transactions (e.g. deep discounting) and are kept in the dataset rather than treated as errors.
- The five-year window (2021–2025) is treated as a sufficiently complete period for trend and seasonality analysis.
- Without a product-to-order key, product-level revenue attribution is out of scope for this version of the project (see Section 5 and Section 16).

14. Expected Deliverables

14.1 Documentation

- This Business Requirements Document
- Data dictionary (Section 7)
- Data-cleaning and data-quality notes (Section 8)
- SQL script(s) with inline comments
- Project README summarizing setup and how to reproduce the analysis

14.2 Database Layer

- Populated database (customers, products, orders tables)
- Data validation queries
- Analytical SQL queries answering the KPIs in Section 9

14.3 Power BI Layer

- Data model connecting customers and orders
- DAX measures for each KPI in Section 9
- The six dashboard pages described in Section 12
- Working slicers and, where useful, drill-down / drill-through

14.4 Business Output

- A short written summary of key findings and insights
- Recommendations grounded in the data (e.g. where returns or negative-profit orders concentrate)
- Documented limitations (Section 5, Section 13)
- Suggested future improvements (Section 16)

15. Success Criteria

- The database is correctly structured and loads all three source files without data loss.
- Data quality findings from Section 8 are documented and explicitly handled in SQL.
- SQL queries produce results that reconcile with the Power BI KPIs.
- Dashboard filters and drill-downs work correctly across all six pages.

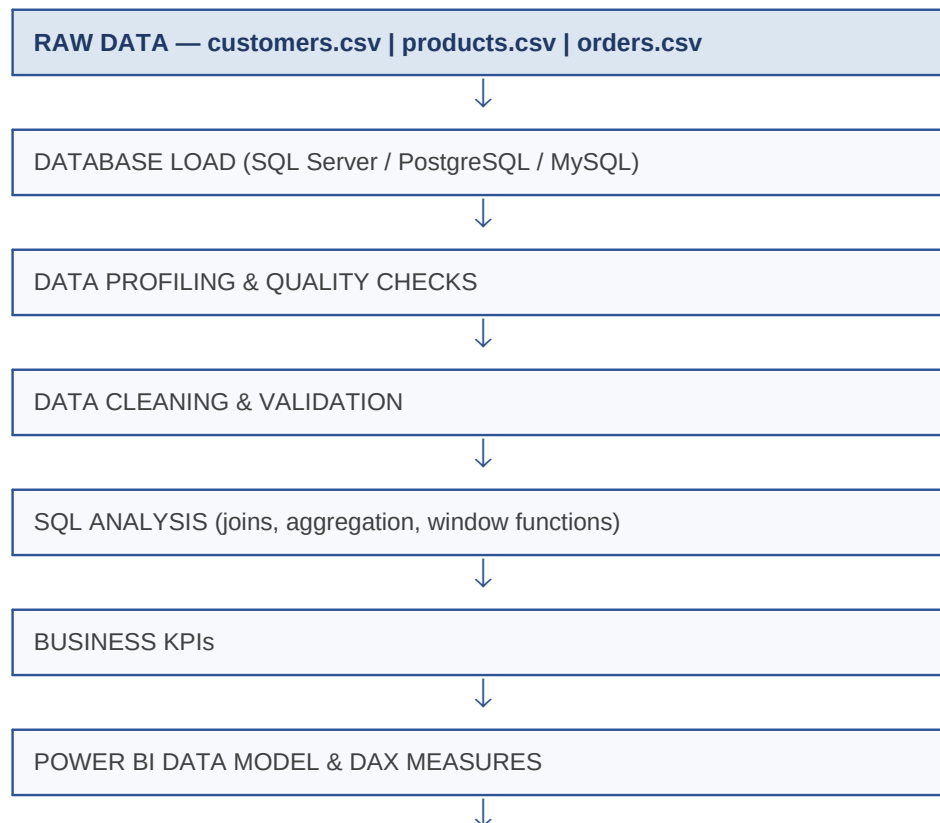
- Every business question listed in Section 2 can be answered directly from the dashboard.
- The write-up clearly explains the analytical process, the key findings, and the known limitations (especially the missing product-order link).

16. Future Enhancements

- Add a synthetic order-line-item table (order_id, product_id, quantity, price) to unlock true product- and category-level revenue analysis.
- Sales forecasting (e.g. simple time-series models on the monthly revenue trend).
- Customer churn prediction using customer_order_count, recency, and lifetime value.
- A full RFM segmentation model with documented, reproducible scoring rules.
- Marketing attribution modeling beyond single-channel counts.
- Automated data refresh and a scheduled dashboard refresh.
- Cohort analysis of customers by acquisition month.

17. Project Architecture

The diagram below summarizes the end-to-end workflow, from raw CSV files through to the final dashboard and business write-up.



DASHBOARDS — Executive | Sales | Customers | Operations | Marketing | Catalog



BUSINESS INSIGHTS & RECOMMENDATIONS

18. Project Outcome

The finished project will be a small but complete analytics portfolio piece: a cleaned, validated database; a set of documented SQL analyses; and an interactive dashboard that turns 138,116 raw order records into clear, decision-ready answers about sales, operations, marketing, and customer behavior — along with an honest account of what the data can and cannot yet answer.

Raw CSV Files → Database → SQL Cleaning & Analysis → KPIs → Power BI Model → Dashboards → Business Insights